



How To Harvest Electricity From A Greenhouse Rooftop

Clear solar cells in greenhouse roofs could generate electricity while plants grow below



Rathindra Nath Biswas is a 1964-batch chemical engineering graduate from Jadavpur University, Calcutta. He was awarded a certificate for designing Benzol Plant by Giprokoks, USSR, and Certificate of Honour by Indian Institute of Metals. He has published 35 research papers in various journals. He retired from service as head - MECON, Durgapur

Because the visible and the infrared (IR) regions of light take up about 47 and 51 per cent of energy in the solar spectrum, respectively, utilising visible light for plant growth and IR light for power generation is extremely exciting. Fig. 1 shows such a concept. IR-absorbing transparent organic photovoltaics (TOPV), with localised IR absorption and visible light transmittance, would be promising materials for this purpose.

These TOPV would also be flexible enough to be spread easily over contours of a greenhouse roof. Using solar cells for both power generation and farming is a clever idea. In the near future, greenhouses might become more popular for farming. As climate change makes growing conditions less predictable, greenhouses could give farmers more control.

Guidelines for selecting solar cells

While implementing this idea commercially, certain factors must be taken into consideration. Solar cells are devices that turn sunlight into electricity. This technology offers a more Earth-friendly way to produce power than burning coal and other fossil fuels. But panels of solar cells need a lot of open, sunny spaces to harvest that sunlight. Where space is limited, people might have to choose between solar panels and fields to grow food. But what if they want to produce crops and electricity at the same time?

The solution is installation of clear solar cells. However, whether enough light can pass through solar cells is not the only factor an engineer needs to think about; there is also efficiency. Common rooftop silicon solar cells, which absorb visible light,

have an efficiency of about eighteen per cent. This means, these produce eighteen watts for every hundred watts of sunlight they absorb.

Transparent flexible organic carbon-based solar cells are most suited for this purpose. These have ten per cent efficiency and are commercially-viable as these are inexpensive.

Sunlight is made up of many colours, or wavelengths, of light. Colours we see from violet to red are called visible light. Their wavelengths range from 400 to 700 nanometres. A nanometre is a billionth of a metre. TOPVs let visible light pass through, which makes them look clear, but at the same time they absorb some IR light, which has wavelengths between 700 nanometres to 1 millimetre to generate electricity.

About half the sunlight that comes from the sun is infrared (IR). People cannot see it, but some animals like snakes and bats can sense IR light. Plants do not need IR light. In fact, they need a tiny range of visible light to grow. Most plants need red and blue lights but reflect green light. That is why most plants look green. TOPVs use IR light for electricity generation and allow visible light to pass through for plants to grow. As entry of IR light is restricted, interiors of a greenhouse are cooler, which is an added advantage. Hence, TOPVs are a preferred option.

Advantages of a greenhouse

While greenhouses have existed for more than 150 years in various parts of the world, in India greenhouses started from 1988 onwards. About 103 big projects at an estimated cost of more than ten billion rupees have been implemented around Ladakh, north-eastern states, Pune, Bengaluru, Hyderabad, New Delhi and so on.

Some advantages of greenhouses are:

- Yield may be ten to twelve times more than that of outdoor cultivation depending

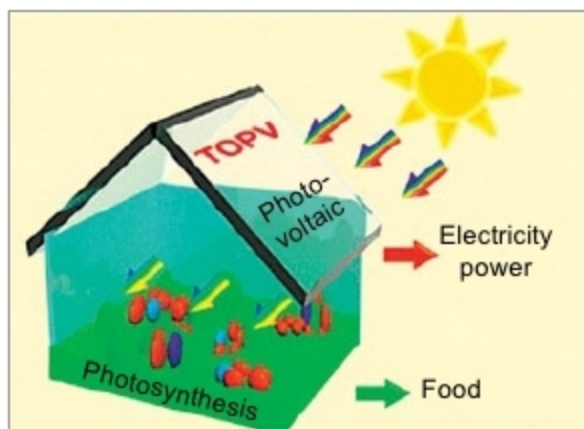


Fig. 1: Schematic arrangement of a greenhouse harvesting electricity